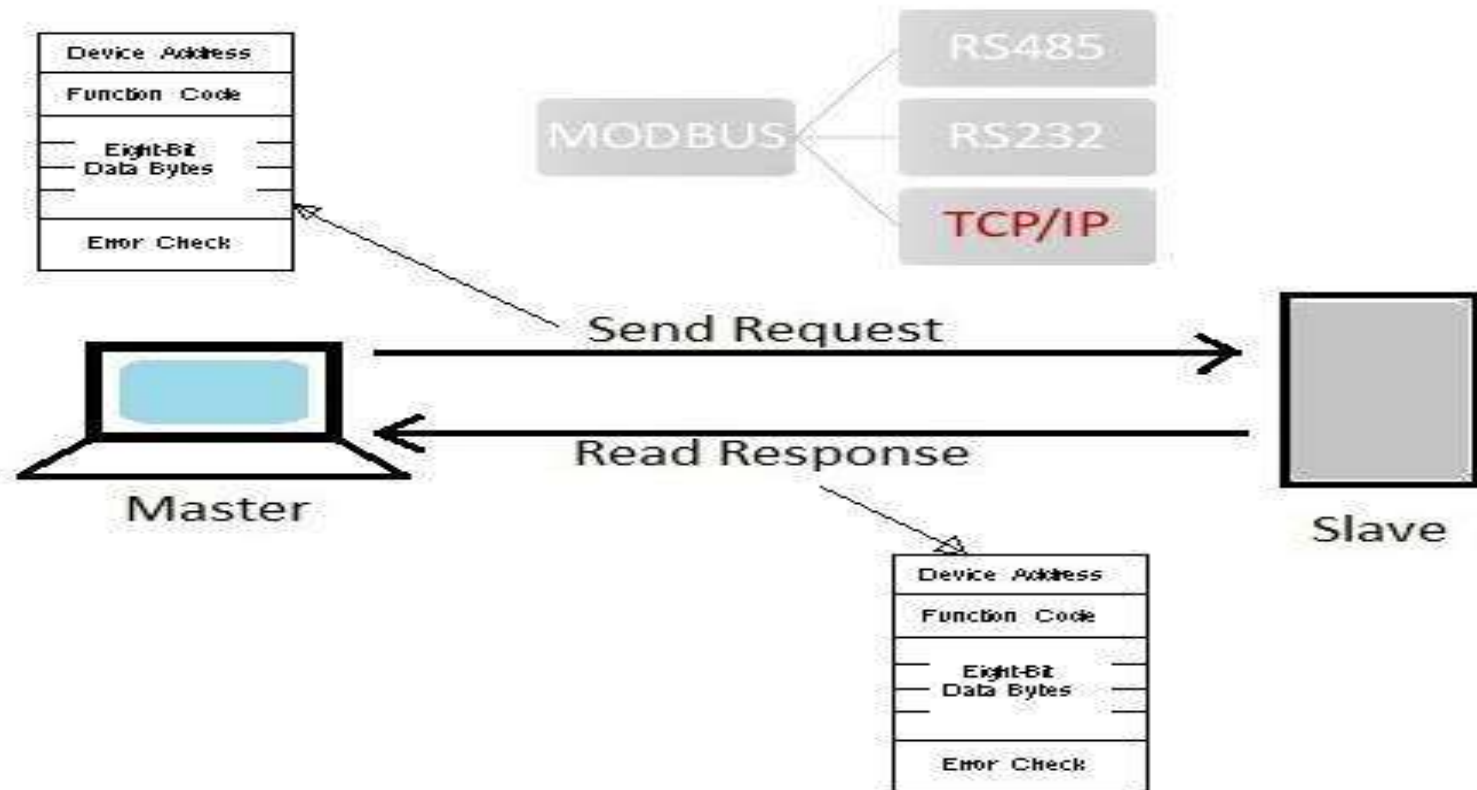


What is Modbus?

- *Application Layer Protocol and it's open source*
- *Developed by Modicon*
- *Transmitting information over serial lines between electronic devices. The device requesting the information is called the Modbus Master and the devices supplying information are Modbus Slaves. In a standard Modbus network, there is one Master and up to 247 Slaves, each with a unique Slave Address from 1 to 247. The Master can also write information to the Slaves.*

Modbus Architecture?



Network Interface/ Medium?

- *RS232, RS485, Ethernet are the Communication Interface/medium used in Modbus Protocol*
- *Modbus is the Language being Spoken*
- *RS232, RS485 and Ethernet are the medium such as Phone, Fax, Letter etc..*
- *Different Methods of Communicating the same core Language between two devices.*

Specification/Settings

- **Device Address/ Slave ID** - Unique Address assigned for each devices in the Modbus Network. Programable for 1 to 247 Devices.
- **Baud Rate** speed of communication in bits/sec. It must be identical for all devices on the Network.
- **Start Bit and Stop Bits** - When a transmitter starts to send character first sit will send start bit followed by character and end by stop bit.
- **Different Methods of Communicating the same core Language between two devices.**

Registers and Tables

- ***Tables are Charts used to define the Registers.***
- ***Baud Rate*** speed of communication in bits/sec. It must be identical for all devices on the Network.
- ***Start Bit and Stop Bits*** - When a transmitter starts to send character first bit will send start bit followed by character and end by stop bit.
- ***Different Methods of Communicating the same core Language between two devices.***

RS485?

- *RS 485 is EIA standard allows high data rates over Long Distance (upto 4000 ft)*
- *Can be Full Duplex (4 wire TX+, TX-, RX+, RX- along with Ground) or Half Duplex (2 wire (TX,RX with Ground))*
- *Multidrop (Multiple devices on the same Bus).*



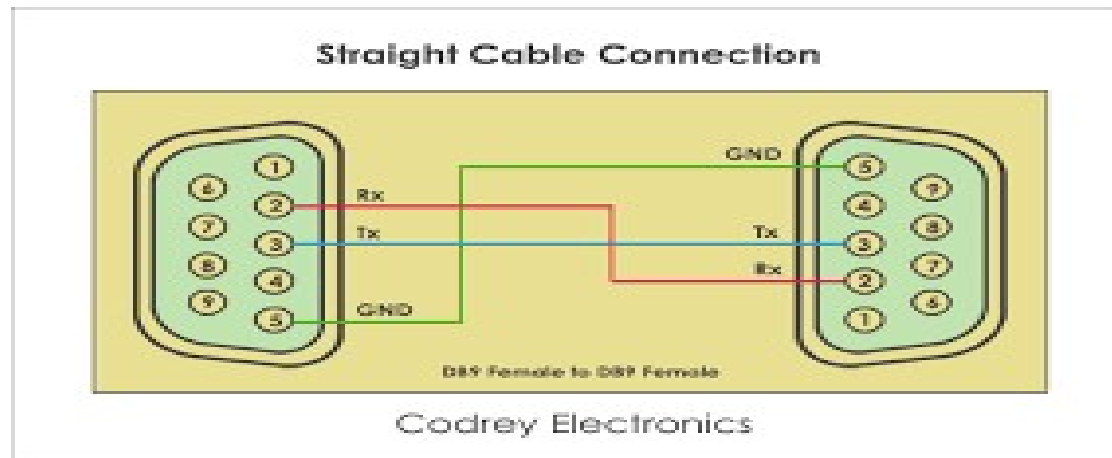
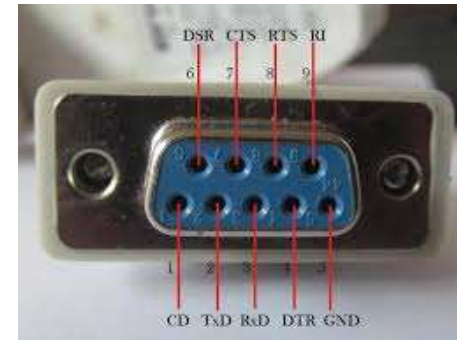
RS485 4 Wire



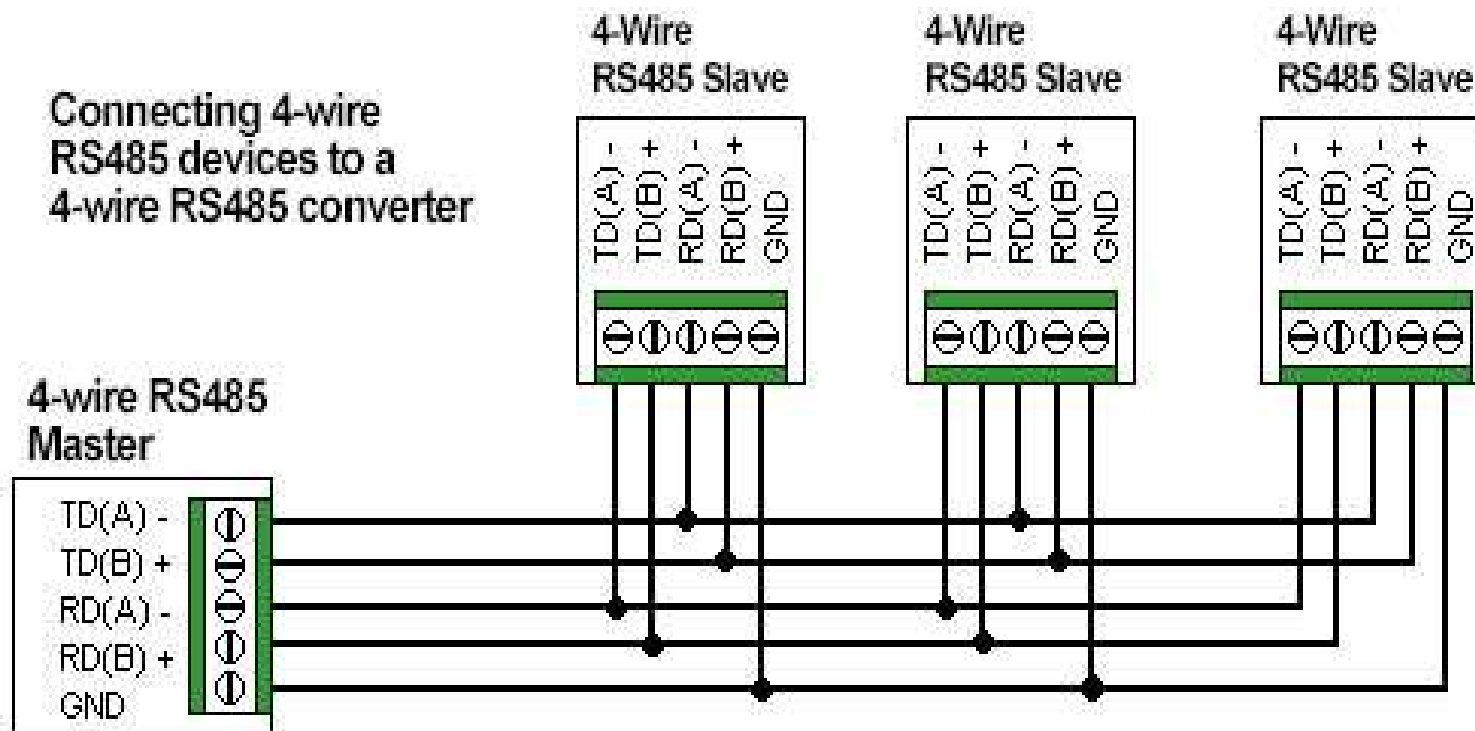
RS485 2 Wire

RS232?

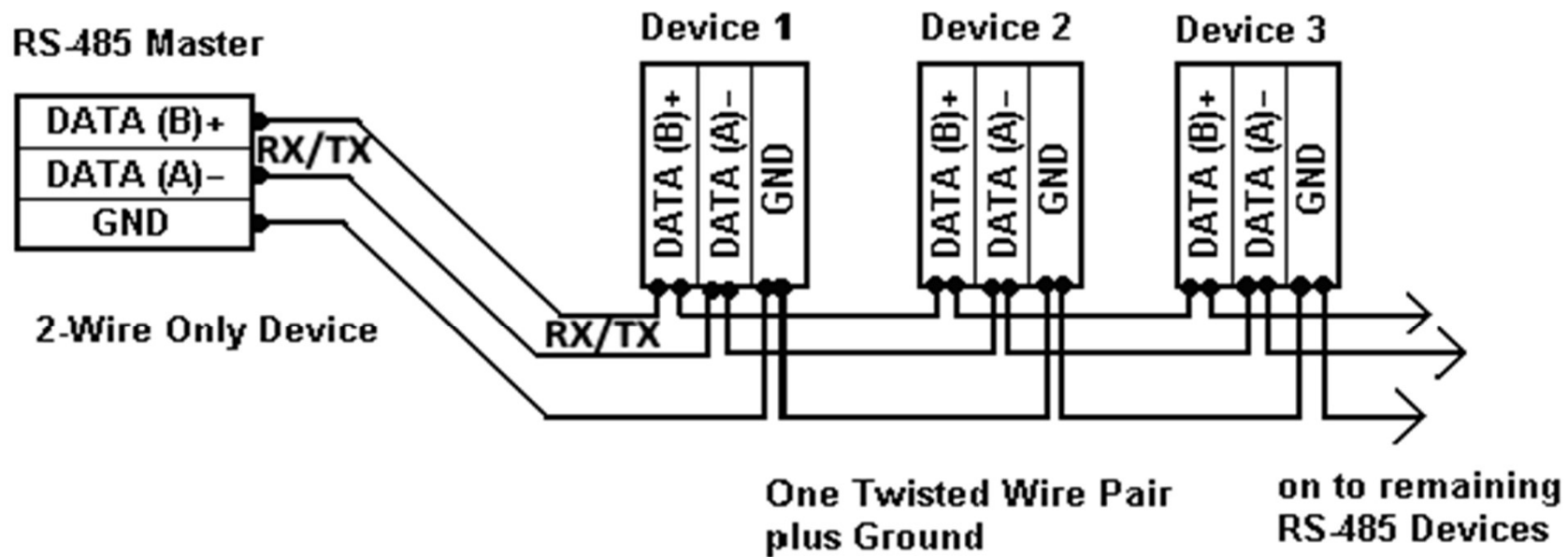
- *RS 232 is Bidirectional Point to Point Link*
- *Max Speed 20Kbit/s.*
- *Upto 50 feet*
- *Usually 9 Pin serial Port*



RS485 4 Wire



RS485 2 Wire



RS485 2 Wire to 4 Wire

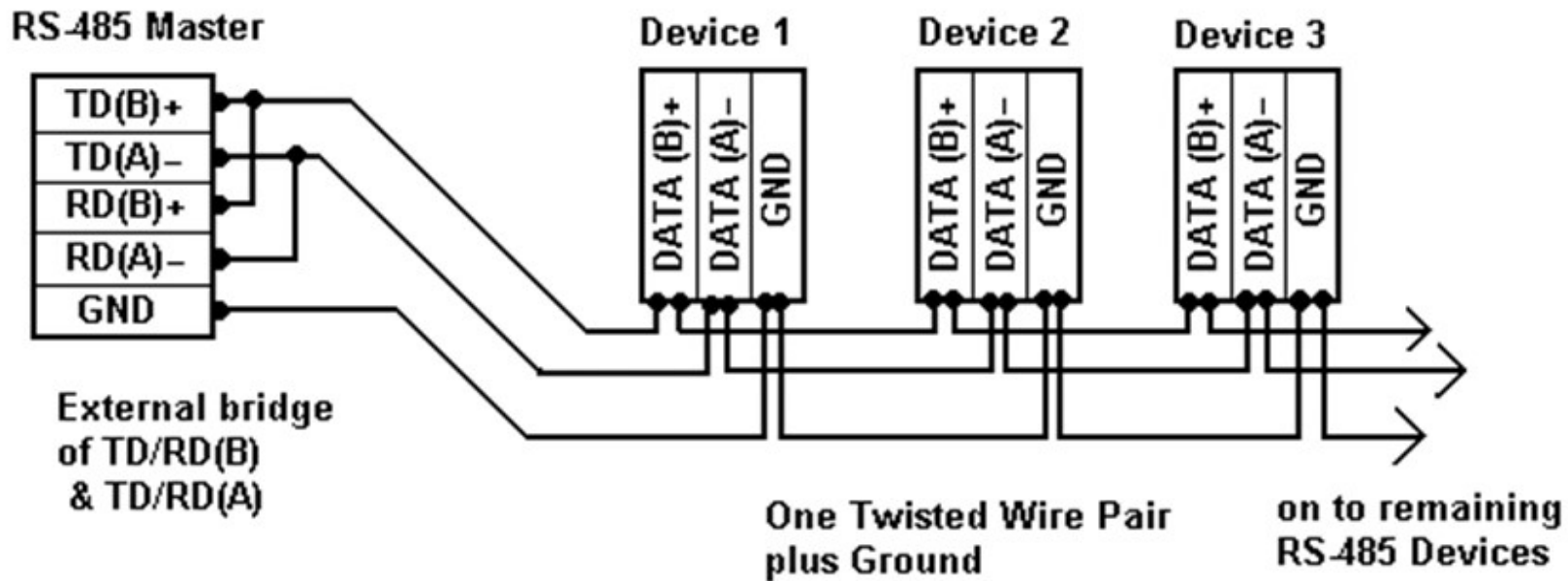


Fig. 3
2-Wire RS-485 Connections

Types

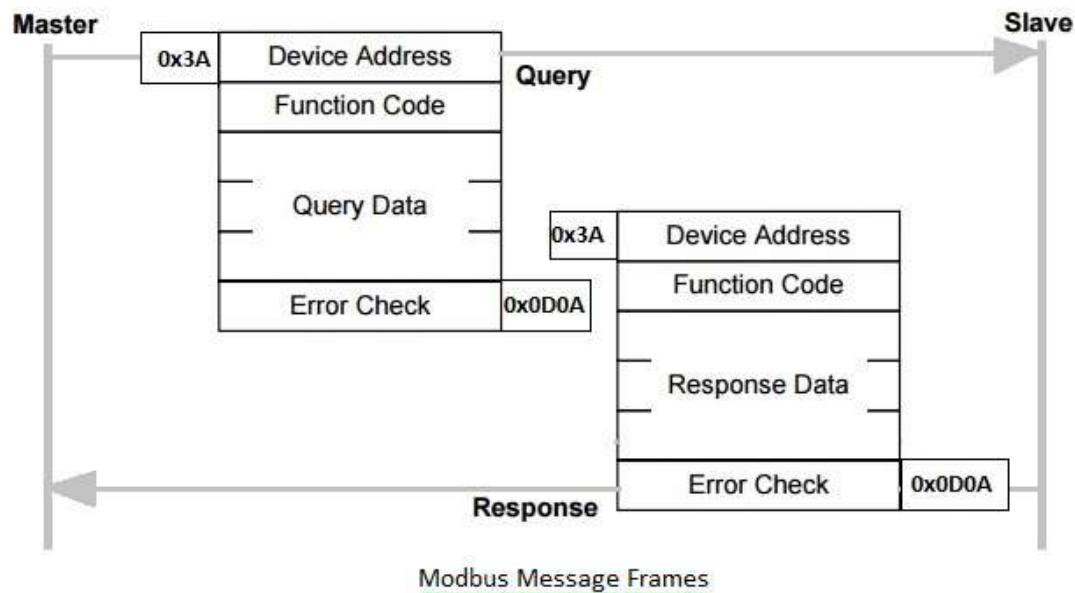
- *Modbus RTU.*
- *Modbus TCP/IP.*
- *Modbus ASCII (American Standard Code for Information Exchange).*

ASCII

- *Each 8 Bit message sent as two ASCII Characters.*
- The main advantage of this mode is that it allows time intervals of up to one second to occur between characters without causing an error.
- ***Modbus ASCII (American Standard Code for Information Exchange).***

Modbus RTU

- *Based on Master- Slave Architecture.*
- Interfaces – RS232, RS485.



Modbus RTU

- **Device Address/ Slave ID** - Unique Address assigned for each devices in the Modbus Network. Programable for 1 to 247 Devices.
- **Function Code** The second byte sent by the Master is the Function Code. It tells the slave which table to access and whether to read from or write to the table.

FC- 01	Read output Coil Status	This command is requesting the ON/OFF status of discrete coils
FC- 02	Read input Coil Status	Read of one or more input coil status
FC- 03	Read Holding Registers	Requesting the content of Analog output Holding registers
FC-04	Read Analog input Registers	Requesting the content of Analog Input registers
FC- 05	Write output Coils	ON/OFF Command for output coils of slave device
FC- 06	Write Holding Registers	write Analog output Holding registers
FC- 15	Write multiple output Coils	write multiple ON/OFF Command for output coils of slave device (Sequence of operation)
FC- 16	Write multiple Holding Registers	write multiple Analog output Holding registers

Modbus RTU

- *Based on Master- Slave Architecture.*
- Interfaces – RS232, RS485.

Modbus RTU Frame Structure

- *Query from Master for Read Coil Status from Slave Device.*

11 01 0013 0025 0E84

11 - *Device/Slave address in Hexa decimal*

01 - *Read Coil Status*

0013 - *Data address of the first coil to Read (19+Offset1 = 20)*

0025 - *Total no of Coil Requested (37 (from 20 to 56)*

0E84 - *CRC (Cyclic redundancy check)*

- *Response from Slave.*

11 01 05 CD6BB20E1B 45E6

11 - *Device/Slave address in Hexa decimal*

01 - *Read Coil Status*

05 - *The number of data bytes to follow (37 Coils / 8 bits per byte = 5 bytes)*

CD - *Coils 27 - 20 (1100 1101)*

6B - *Coils 35 - 28 (0110 1011)*

B2 - *Coils 43 - 36 (1011 0010)*

0E - *Coils 51 - 44 (0000 1110)*

1B - *3 space holders & Coils 56 - 52 (0001 1011)*

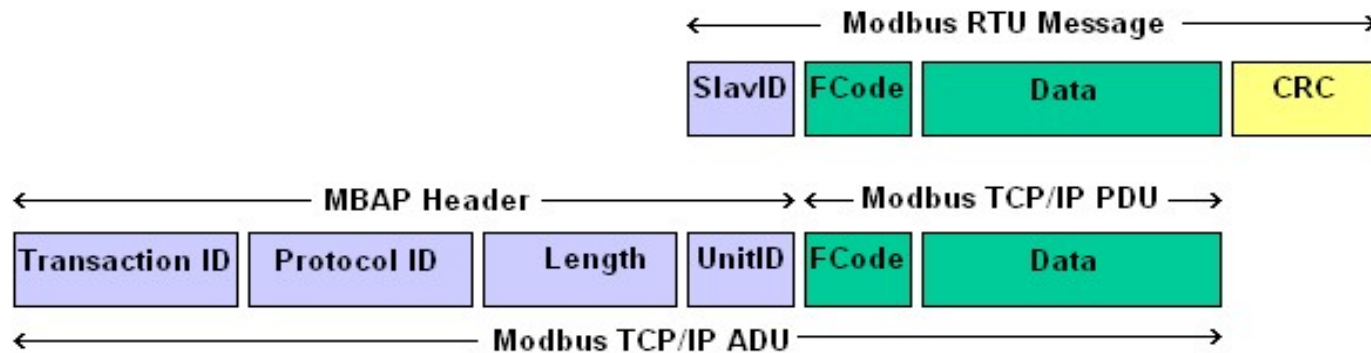
49AD *CRC (Cyclic redundancy check)*

Modbus RTU Frame Structure

- *Query from Master for Read output Holding registers (FC – 03).*
 11 03 006B 0003 7687
 11 - Device/Slave address in Hexa decimal
 03 – Function Code 3 (read Analog Output Holding Registers)
 006B - Data address of the first register requested (107+Offset40001 = 40108)
 0003 - Total no of Registers Requested (3 (from 40108 to 40110)
 7687 – CRC (Cyclic redundancy check)
- *Response from Slave.*
 11 03 06 AE41 5652 4340 49AD
 11 - Device/Slave address in Hexa decimal
 03 – Function Code 3 (read Analog Output Holding Registers)
 06 - The number of data bytes to follow (3 registers x 2 bytes each = 6 bytes)
 AE41 - The contents of register 40108
 5652 - The contents of register 40109
 4340 - The contents of register 40110
 45E6 CRC (Cyclic redundancy check)

Modbus TCP/IP

- *Modbus RTU message transmitted with a TCP/IP wrapper and sent over a network instead of serial lines.*
- *Running on Ethernet over TCP/IP interface.*



Modbus TCP/IP Frame Structure

- *Query from Master for Read Analog output Holding Register from Slave Device.*

0001 0000 0006 11 03 006B 0003

0001 - *Transaction Identifier*

0000 - *Protocol Identifier*

0006 - *Message Length (6 Bytes to follow)*

11 - *Unit Identifier 11 (17 Hex)*

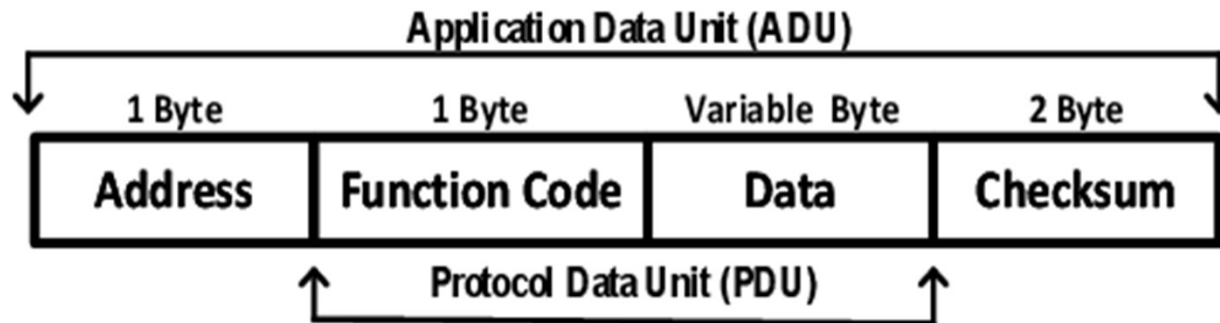
03 - *Function Code (Read Analog output Holding Register)*

006B - *The data address of the first register requested (40108-40001 = 107 Hex- 006B)*

0003 - *The Total no of Registers Requested (3)*

ADU/PDU

- *ADU – Application Data Unit is an envelope containing a message, the PDU (Protocol Data Unit) is the message itself.*



Modbus Data Type

UINT 16	16-bit unsigned integer (1 word)	Range: 0 to 65535
INT 16	16-bit signed integer (1 word)	Range: -32768 to 32767
INT32	32-bit signed integer (2 words)	Range: -2 147 483 648...+2 147 483 647
UINT32	32-bit unsigned integer (2 words)	Range: 0...4 294 967 295
Float32	32-bit value (2 words)	Range -3.4028E+38... +3.4028E+38
ASCII	8-bit alphanumeric character	
BITMAP	16-bit field (1 word)	